

Differentially Private Steering for Large Language Model Alignment

Hypothesis: It is possible to effectively align LLM behavior using activation editing while safeguarding the privacy of the alignment dataset through Differential Privacy (DP). By adding calibrated Gaussian noise to steering vectors computed from private demonstrations, we can create a differentially private alignment method, PSA, that allows an LLM to answer user queries with formal privacy guarantees with respect to the private data, while minimally affecting its alignment, text generation, and general capabilities.

Related Work: Recent research has focused on aligning LLMs using techniques like RLHF, instruction tuning, and activation editing. Activation editing, which steers LLM behavior by modifying activations, is a lightweight alternative to fine-tuning. However, while the privacy risks of fine-tuning and in-context learning have been studied, the privacy implications of using private demonstration data for activation editing remain an unexplored research gap. This work aims to address this gap by developing the first activation editing method that adheres to formal Differential Privacy guarantees.

Abstract: [Content omitted for brevity]

Experiments:

Description: To evaluate the alignment performance of the proposed PSA algorithm against non-private methods and zero-shot baselines on multiple-choice question benchmarks.

Method: The PSA algorithm will be applied to several open-source LLMs (LLaMa-2 7B, Mistral-v0.3 7B, Gemma-2 2B, Qwen-2.5 7B). Performance will be evaluated on seven alignment-relevant benchmark datasets from Anthropic's 'Advanced AI Risk' collection (Sycophancy, Hallucination, Refusal, Myopic Reward, Survival Instinct, Corrigibility, Coordination). Baselines for comparison will be non-private Mean Steering, PCA Steering (In-Context Vectors), and the zero-shot performance of the LLMs

Metrics: Accuracy in choosing the correct behavioral option in multiple-choice questions.

Description: To evaluate the quality of open-ended text generation and instruction-following capabilities of LLMs steered with PSA.

Method: The Llama-2-Chat 7B model will be steered using PSA, Mean Steering, and PCA Steering. The models will be prompted with open-ended questions from the test set of Rimsky et al. (2024).

Metrics: Qualitative scores from GPT-4 used as an LLM evaluator to judge the behavior and quality of the generated text.

Description: To investigate the scaling behavior of PSA and the privacy-utility tradeoff across LLMs of varying sizes

Method: The performance of PSA and non-private mean steering will be tracked across the Qwen-2.5 series of models with varying sizes (0.5B, 1.5B, 3B, 7B) on the Refusal dataset.

Metrics: MCQ Accuracy, GPT-4 Evaluation score for text generation, MMLU Performance

Description: To empirically evaluate the privacy leakage of activation editing and audit the effectiveness of PSA using a novel Membership Inference Attack (MIA).

Method: A new MIA will be developed, tailored for activation editing. This attack involves inserting 'canary' demonstrations into the alignment dataset and checking if the steered LLM prefers the canary information during inference. The attack will be run against both non-privately steered (Mean Steer) and PSA-steered LLMs (Llama-2-7B, Qwen-2.5-7B) using the Hallucination dataset.

Metrics: False Positive Rate (FPR), False Negative Rate (FNR), Empirical ϵ (epsilon) privacy parameter

Risk Factor:

While mean steering is a straightforward and popular method to steer LLMs, recently more sophisticated methods have been proposed.

Another limitation of our work is that we only consider alignment on a specific set of axes and in a monolingual setting.

Meta ICLR review: The submission proposes Private Steering for LLM Alignment (PSA), an algorithm that performs activation editing with a differential privacy (DP) guarantee on a sensitive dataset. The submission empirically evaluates the performance of this algorithm, comparing it to non-DP activation editing algorithms as well as the zero-shot capabilities of 3 state-of-the-art LLMs of different sizes on standard alignment benchmarks. The submission also proposes a membership inference attack (MIA) for this setting, and shows that this attack can be successful when non-DP steering is performed. Overall, the submission studies a timely problem, proposes a simple DP algorithm, and performs a thorough evaluation of its performance.

Mean ICLR Soundness Score

3.25/4.0

👍

🌟 Claude Opus 4.6

GPT-5.4 Thinking

🌟 Gemini 3.1 Pro

👍

👍

👍